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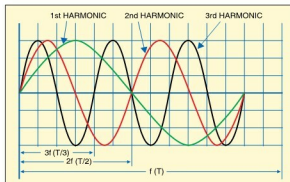


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Q1. What are harmonics? How are these created? What are their effects?

S.N. Syed Mohamed Buhari
 Tirunelveli

A1. Harmonics are an integer multiple of the fundamental frequency. For example, if the fundamental frequency is f , the harmonics



Harmonics f , $2f$, $3f$

have frequencies f , $2f$, $3f$, $4f$, etc. These harmonics are called first, second, third, fourth, etc. For example, if the fundamental frequency is 400 Hz, the frequencies of the harmonics are 400 Hz, 800 Hz, 1200 Hz, 1600 Hz and so on. Harmonics are shown here in the figure.

Harmonics have the property that these are all periodic at the fundamental frequency. These are not overtones. However, there is a relationship as the first harmonic is known as the fundamental tone, second harmonic as the first overtone, third harmonic as the second overtone, fourth harmonic as the third overtone and so on.

Harmonics are generated when a signal is processed by a non-linear

system—a system that has non-linear input-output relationship. In general, the output $r(t)$ of a linear system for an input $g(t)$ can be approximated in the polynomial form as:

$$r(t) = a_1 g(t) + a_2 g^2(t) + a_3 g^3(t) + \dots$$

where a_1 , a_2 , and a_3 are constants.

Thus for a signal $g(t) = \cos 2\pi f_0 t$, f_0 being the input frequency, the output

will have harmonics f_0 , $2f_0$, $3f_0$, etc. The output of a non-linear system will have frequencies that were not present in the input signal—a case of non-linear distortion. If the input signal is sum of multitones, say, f_1 and f_2 , the output will have cross-product terms $f_2 - f_1$, $f_2 + f_1$, $f_2 - 2f_1$, $2f_2 - f_1$ besides harmonics of f_1 and f_2 . These additional sum

and difference frequencies indicate additional distortions that are designated by intermodulation distortion.

Thus generation of harmonics leads to harmonic distortion—an undesirable thing in amplifier operation. Note that the same cross-product term can be exploited to achieve modulation in communication systems.

Q2. What's the pin configuration of SD memory cards that are used in mobile phones and interfacing with the USB port?

James Rajavarman
 Through e-mail

A2. Secure Digital (SD) cards are removable flash memory cards. These are commonly used in

TABLE I
Pinout Details of microSD Cards in SD and SPI Modes

Pin No.	Pin name	SD signal function SD mode	SPI signal function SPI mode
1	DAT2/NC	Data bit 2	No connect
2	CD/DAT3	Card detect/Data bit 3	Chip select
3	CMD/DI	Command line	Master out/Slave in (MOSI)
4	Vdd	Supply voltage 2.7V/3.6V	Supply voltage 2.7V/3.6V
5	CLK	Clock	Clock
6	Vss	Ground	Ground
7	DAT0/DO	Data bit 0	Master in/Slave out (MISO)
8	DAT1/RSV	Data bit 1	Reserved